

Comparing Readability Formulas for Chatbot and Online Content Clarity

Background: Readability Formulas and Their Purpose

Readability formulas are objective, mathematical tools that estimate how easy a text is to read. They produce numerical scores corresponding to the reading skill (often a U.S. school grade level) needed for comprehension ¹. Historically defined as “the ease of understanding or comprehension due to the style of writing” ², these formulas rely on surface textual features – typically sentence length and word difficulty (e.g. syllable count) – to gauge clarity. Over the past century, over 200 different readability formulas have been developed ³, many of which remain in use today for evaluating everything from school books to web articles.

Common Readability Metrics: Several classic formulas are frequently applied in research and practice ⁴ ⁵:

- **Flesch Reading Ease (FRE):** Uses average sentence length and syllables per word to yield a score from 0–100 (higher = easier to read) ⁴. For example, a text scoring 60–70 is considered reasonably easy for the average adult.
- **Flesch-Kincaid Grade Level (FKGL):** Uses similar inputs (sentence length and syllables) but outputs an approximate U.S. school grade level (e.g. an FKGL of 11.2 $\approx$ 11th-grade level) ⁶.
- **Gunning Fog Index:** Estimates the grade level needed to read a text, factoring in complex words (typically those ≥ 3 syllables). It assumes that more multisyllabic “difficult” words per sentence increase reading difficulty ⁵.
- **SMOG Index (Simple Measure of Gobbledygook):** Predicts the years of education required to fully comprehend a text, based on the count of polysyllabic words. SMOG was designed to estimate **100% comprehension** of a passage (hence often giving slightly higher grade levels) and has been widely used for health materials ⁷.
- **Coleman-Liau Index:** Uses characters per word and sentences per 100 words to calculate a U.S. grade level, not relying on syllable counting (useful for automated analysis).
- **Automated Readability Index (ARI):** Similar to Coleman-Liau, ARI uses characters and words per sentence to produce a grade estimate.
- **Dale-Chall Formula:** Looks at the proportion of “difficult” words (not on a list of 3,000 common words) in a text, combined with sentence length, to assign a grade level ⁸. Texts with more uncommon vocabulary score as requiring a higher reading level.

These formulas provide quick, **objective scores** to assess whether content matches the reading ability of a target audience. For instance, public-facing materials in the U.S. are often recommended to aim for around a 6th–8th grade level for broad understandability ⁹.

Limitations: Traditional readability metrics are valued for their transparency and ease of use, but they have well-known limitations. They focus on quantifiable text features (length of words/sentences), **ignoring**

deeper linguistic factors like vocabulary difficulty, syntax complexity, or reader background knowledge ¹⁰ ¹¹ . For example, “*banana*” is a three-syllable word that these formulas deem complex, even though it’s easily understood (a critique often leveled at Gunning Fog’s syllable rule). Similarly, domain-specific jargon or poor text organization can hurt comprehension despite short words or sentences – factors outside the scope of classic formulas ¹⁰ . Because of this, readability scores sometimes **misjudge actual comprehension**. Researchers have noted that formulas like Flesch–Kincaid “fail to accurately measure text difficulty” on their own ¹⁰ . This has led to efforts to improve or supplement traditional metrics with more linguistically rich approaches (e.g. **Coh-Metrix** tools, or machine learning models that account for semantics and coherence ¹² ¹³). Nonetheless, the old formulas remain popular due to their simplicity, proven track record, and interpretability ¹⁴ ¹⁵ .

Recent Comparative Studies of Readability Metrics

To understand which readability scores are most effective for the “average American reader,” recent research has evaluated how well these formulas actually reflect comprehension difficulty. A notable study by Lee *et al.* (2023) revisited the classic formulas in modern contexts ¹⁶ . They **recalibrated five traditional indices** – Flesch–Kincaid, Gunning Fog, SMOG, Coleman–Liau, and ARI – using contemporary U.S. educational texts, and introduced a new composite metric called the **New English Readability Formula (NERF)** ¹⁶ . This work built a tool to cross-compare formula performance on diverse reading materials (from grade-school texts to medical documents). The findings showed that updating the old formulas’ coefficients to today’s language use substantially **improved their accuracy** in predicting text grade levels ¹⁷ . More intriguingly, the new NERF formula (which incorporates a broader range of lexical and syntactic features) outperformed all the individual traditional formulas in aligning with human-judged readability levels ¹⁷ ¹⁸ . In other words, NERF was better at matching what a panel of readers with average education would consider appropriately easy or hard text. Recalibration helped the older metrics, but NERF still achieved the closest agreement with actual difficulty rankings ¹⁷ . Lee *et al.*’s work serves as a **comprehensive update and comparison** of readability metrics, highlighting that while formulas like FKGL or SMOG can be tuned to modern text, a multi-factor approach (accounting for word familiarity, age-of-acquisition, etc.) can yield a more reliable readability assessment ¹⁹ ²⁰ .

Despite such improvements, **other studies cast doubt on the overall effectiveness** of readability formulas in predicting true comprehension for the average reader. Gruteke Klein *et al.* (2025) took a novel approach by evaluating several readability scoring methods against *real-time reading ease* data ²¹ . They used eye-tracking measures (reading speed, eye fixations) as an objective gauge of how easily people read texts, and compared how well classic formulas, modern NLP models, and even large language models (LLMs) could predict these measures. The verdict: **none** of the existing readability methods correlated strongly with actual reading ease ²² . Traditional formulas, machine-learning systems, and AI-based tools were all **poor predictors of real reader effort** when tested with eye-tracking data ²² . In fact, simple lexical properties from psycholinguistics (like word frequency or length) often outperformed the sophisticated metrics in explaining variance in reading ease ²² . This suggests a fundamental limitation: current readability scores, even when mathematically refined, might be missing key cognitive factors that determine how “readable” a text truly is (such as narrative flow, cohesion, or readers’ prior knowledge).

Consistent with that, an evaluation by Ahmed *et al.* (2025) examined how readability formulas relate to **actual comprehension and perceived difficulty** of health information. In their study, participants read health texts (in either text or audio form) and answered understanding questions, while various formula scores (Flesch, Fog, SMOG, Dale–Chall, etc.) were calculated for those texts ²³ ²⁴ . The results showed a

telling disparity: readability scores had **no meaningful correlation with readers' quiz comprehension or information retention** for the materials, meaning a lower grade-level score didn't guarantee better actual understanding ²⁵ ²⁶ . However, the formulas did correlate moderately with **perceived difficulty** – readers' subjective sense of how hard a text was ²⁵ ²⁶ . For instance, the SMOG index showed a positive correlation (~0.33) with readers' *perceived* difficulty ratings ²⁵ , and even had a **notable (though modest) correlation with retention** of information (SMOG was the one metric that slightly related to how well people remembered content) ²⁶ . Overall, though, Ahmed et al. conclude that traditional readability formulas **"failed to explain actual difficulty"** – texts that the formulas labeled as "easy" didn't consistently yield better comprehension scores ²⁶ . This reinforces the idea that while grade-level formulas are handy indicators (and can predict if readers will *feel* a text is complex), they are imperfect proxies for true understanding.

In summary, recent research suggests that **no single readability formula is infallible**. Simpler metrics (word/sentence length based) capture some aspects of difficulty and are convenient for benchmarking content against education levels. For example, SMOG is often favored in health communication because it's straightforward and was found to give consistent results across many consumer texts ⁷ . One review noted SMOG's "high utility" and more consistent scoring compared to other indices in consumer health materials ⁷ . Yet, even SMOG can agree with other formulas in some cases and diverge in others ²⁷ . The consensus in academia is that readability formulas are **useful but limited** tools: they should be supplemented with qualitative judgment or newer AI-based analysis when high clarity is critical ¹² ¹³ . Researchers are actively exploring improved models – from **recalibrated equations** (like NERF) to **LLM evaluators** – but as of now, identifying content that an "average American reader" can easily comprehend still often requires combining multiple metrics and human insight.

Readability of Chatbot-Generated Content

With the surge of AI chatbots (like OpenAI's ChatGPT or Google's Bard/Gemini) producing content, scholars have started examining how *readable* these AI-generated responses are, and which formulas are best suited to evaluate them. Recent studies typically apply a **battery of standard readability indices** to chatbot outputs, the same way they've been used on websites or pamphlets ²⁸ . For instance, Bükér *et al.* (2025) assessed the answers given by three popular large-language-model chatbots – **ChatGPT-3.5**, **Microsoft's Bing Copilot**, and **Google's Gemini** – to common patient questions about a dental procedure ²⁹ . They used **seven different formulas** (Flesch Reading Ease, FK Grade, SMOG, Gunning Fog, Linsear Write, Coleman-Liau, and ARI) to gauge the reading level of each response ²⁸ . All three chatbots' answers scored *far above* the recommended 6th-grade level for patient materials – in fact, the content was suited to roughly a **10th-grade (or higher) reading level** on average ³⁰ . In practical terms, this means the *average American adult* (about 8th- to 9th-grade reading skill) might find the chatbot explanations somewhat challenging. Among the chatbots, Gemini's answers were the easiest to read: Gemini had **significantly lower grade-level scores** than ChatGPT or Copilot ($p < 0.05$), indicating more accessible phrasing ³⁰ . By contrast, ChatGPT and Copilot's readability scores were statistically similar to each other, both more difficult than Gemini's output ³⁰ . Notably, **none** of the chatbots met the health literacy guideline of 6th-grade level – even Gemini's "simpler" responses overshot that target ³¹ . The authors concluded that while Gemini demonstrated better readability (and even slightly better accuracy/quality in that study), *all three systems need refinement to improve accessibility* of their language for patients ³¹ .

Another study in the medical domain (Özdemir *et al.*, 2025) evaluated responses from **ChatGPT-4**, **Google Gemini**, and **Microsoft Copilot** to questions about children's oral habits ³² ³³ . They found a similar

pattern: ChatGPT's responses, though richly detailed, were written at a **significantly higher reading level** than the other bots' answers ³⁴. In quantitative terms, ChatGPT-4's text had a **lower Flesch Reading Ease score and higher grade level (FKGL)**, meaning it was more complex and difficult, whereas Gemini and Copilot tended to phrase answers more simply ³⁴. Interestingly, this did *not* mean ChatGPT's answers were worse overall – in fact, the study noted that ChatGPT-4 received the highest quality ratings (with about 77% of its answers rated “excellent” in a quality score) and was the most accurate in factual terms ³⁴. The trade-off observed was that Microsoft's Copilot “**excelled in simplicity**” of language, making its content the easiest to read, while ChatGPT excelled in depth and structure ³⁵. The researchers emphasized that **improving readability** should be a priority for these AI tools, so that users can benefit from their information **without struggling to understand the phrasing** ³⁵. In short, current chatbots — even when providing correct and helpful answers — often do so in language above the average user's reading level, and standard readability metrics are instrumental in uncovering this gap.

It's worth noting that readability formulas sometimes **disagree on the exact level** of chatbot text, so using multiple metrics gives a more complete picture. Bahçeci *et al.* (2025) compared an AI chatbot (Microsoft Copilot) against traditional Google search results for common infertility questions, analyzing the text from both sources with several formulas ³⁶. Both the chatbot-generated answers and the Google-sourced information turned out to require an **advanced reading level (above 8th grade)**, consistent with other studies ³⁷. However, only some formulas detected a difference between the two sources – the Automated Readability Index and Flesch-Kincaid Grade showed that the chatbot's answers were *slightly easier* than the Google search results (significantly lower grade scores for the chatbot) ³⁸. In contrast, other indices like Flesch Reading Ease, Gunning Fog, SMOG, and Coleman-Liau *showed no significant difference* between the chatbot and Google content ³⁸. This finding underscores that different formulas, owing to their distinct calculations, can yield **varying appraisals of the same text**. One metric may flag a piece of writing as harder to read while another rates it as moderate, so researchers often report several readability scores in tandem ²⁸. If most indices concur that content is above a certain grade level, one can be more confident that the material is indeed too advanced for the average reader.

Readability in General Online Content

The insights from chatbot studies align closely with readability research on general online content like websites and social media. Numerous audits of health websites, for example, reveal that **online materials are often written at a high-school or college reading level**, exceeding what an average adult might comfortably read ³⁹ ⁴⁰. One review found that patient resources on common conditions (eg. pancreatitis, pulmonary embolism) had a **mean readability over the 10th-grade level** ³⁹. Even sites explicitly aiming for broader audiences sometimes miss the mark: a geriatric health website that intended to use “simple language” was analyzed to be around a 10th-grade level as well ³⁹. In specialized fields, the trend persists – for instance, critical care societies' patient education brochures scored in the “difficult” range (Flesch scores ~50, roughly equating to late high school level) ⁴⁰. Studies across medical specialties (oncology, rheumatology, etc.) consistently find that **most web-based patient education fails to meet the 6th-8th grade guideline** ⁴¹ ⁴². Social media content can vary widely, but when users share technical or scientific information, readability can similarly skew high. The common theme is that **plain language is the exception rather than the norm** on many platforms, which numerous authors have flagged as a barrier to public understanding ⁴³ ⁴⁴. This has driven home the importance of **readability assessment**: by quantifying text complexity, one can identify when rewriting or simplification is needed to make content accessible to a broader audience.

Conclusion

In summary, academic evaluations of readability formulas over the last decade show a nuanced picture. Traditional formulas like Flesch-Kincaid, Gunning Fog, and SMOG remain handy tools to **quantify text clarity**, and they are actively used to evaluate both chatbot outputs and conventional online text ²⁸ ¹. They can highlight when content overshoots the reading ability of an average American (for example, flagging that a chatbot's answer is at a 11th-grade level instead of the recommended 6th-grade level ³⁰ ³¹). Among these classic metrics, **SMOG** often stands out in health literacy research for its reliability and conservative estimates, sometimes providing more consistent results across different texts ⁷. However, no single score is universally "most effective" – each has blind spots, and their effectiveness can depend on the context. Recent research indicates that while readability scores correlate with how *difficult readers think* a text is, they **don't always guarantee comprehension** ²⁵ ²⁶. Advanced approaches (like the NERF formula or AI-based readability evaluators) are being developed to address these gaps ¹⁶ ¹⁷. For now, the best practice is to use readability formulas as a **starting point**: if multiple indices concur that content is above an 8th-grade level, it's a strong signal the text may challenge average readers. From there, writers or chatbot designers can simplify wording and structure to improve understanding. In the end, the goal is not to chase a specific formula's score, but to ensure the **content can be comfortably understood by its intended audience** – something the combined evidence from these papers strongly encourages.

Sources: Recent studies and reviews supporting these findings include Lee *et al.* (2023) ¹⁶ ¹⁷, Gruteke Klein *et al.* (2025) ²², Ahmed *et al.* (2025) ²⁵ ²⁶, Bükér *et al.* (2025) ³⁰ ³¹, Özdemir *et al.* (2025) ³⁴ ³⁵, Bahçeci *et al.* (2025) ³⁸, and a 2024 MDPI review of readability in patient education ⁴⁵ ⁴, among others. Each provides a unique lens on how readability formulas perform and how they are applied to ensure content is **clear and comprehensible** to the general public.

¹ ⁹ ²⁸ ²⁹ ³⁰ ³¹ Readability, accuracy and appropriateness and quality of AI chatbot responses as a patient information source on root canal retreatment: A comparative assessment - ScienceDirect
<https://www.sciencedirect.com/science/article/abs/pii/S1386505625001650>

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¹⁶ ¹⁷ ¹⁸ ¹⁹ ²⁰ Traditional Readability Formulas Compared for English
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²³ ²⁴ ²⁵ ²⁶ Parallel Corpus Analysis of Text and Audio Comprehension to Evaluate Readability Formula Effectiveness: Quantitative Analysis - PMC
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